

Chapter 4 — Audio, video, and tabular annotation

Text and image schemas are only part of the annotation landscape

Learning objectives

- Describe typical audio labels such as transcripts and speaker turns
- Recognize row categorization and outlier flags as tabular annotation patterns

Audio annotation techniques

- Audio annotation supports speech recognition, speaker diarization, and affect analysis
- Typical labels include transcripts, speaker identities, and emotion tags aligned to time
- Transcription converts speech to text for automatic speech recognition training
- Diarization assigns who spoke when in multi-speaker recordings
- Emotion tagging labels affective tone in discrete classes such as happy

Example 4.17 — Transcription on a customer service call

- Example 4.17 — hands-on module
- Example 4.17 shows a call-center transcript task
- Annotators listen to a call and produce a time-aligned transcript used to train automatic
- Alignment quality
- Explore the chapter example module
- View files: ``modules/chapter4/example17/``

Video annotation techniques

- Video annotation combines spatial and temporal labeling and is costly because objects and
- Frame-by-frame labeling annotates objects in successive frames for detection and tracking
- Activity recognition labels clip-level behaviors
- Production programs often separate dense tracks from lighter activity tags to control cost

Example 4.21 — Frame-by-frame labeling on video

- Example 4.21 — hands-on module
- Example 4.21 shows surveillance-style classes
- In surveillance video
- Temporal consistency across frames is a QC concern
- Explore the chapter example module
- View files: ``modules/chapter4/example21/``

Tabular data annotation

- Tabular annotation assigns labels to rows
- Categorization maps rows to discrete classes
- Labeling outliers marks anomalous rows for fraud or quality review when positives are rare

Example 4.24 — Categorization on tabular rows

- Example 4.24 — hands-on module
- Example 4.24 shows customer-value bands
- In a customer database
- The schema must define thresholds and tie-break rules so annotators
- Explore the chapter example module
- View files: ``modules/chapter4/example24/``

Takeaways

- Audio work aligns text, speakers, or affect with time
- Video adds temporal cost to spatial labels and often mixes tracks with activity tags
- Tabular labels include class bands and outlier flags for classification and fraud

Next

- Complete the quiz for this part
- Crowdsourced workflows with their trade-offs